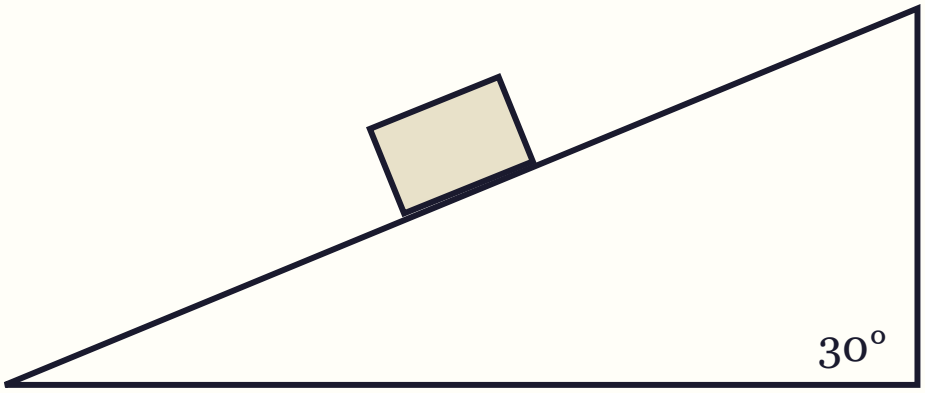
Kinematics

Mistake 1

Block on a rough incline

Coaching DPP · DPP 4 Q12 · Silly calculation

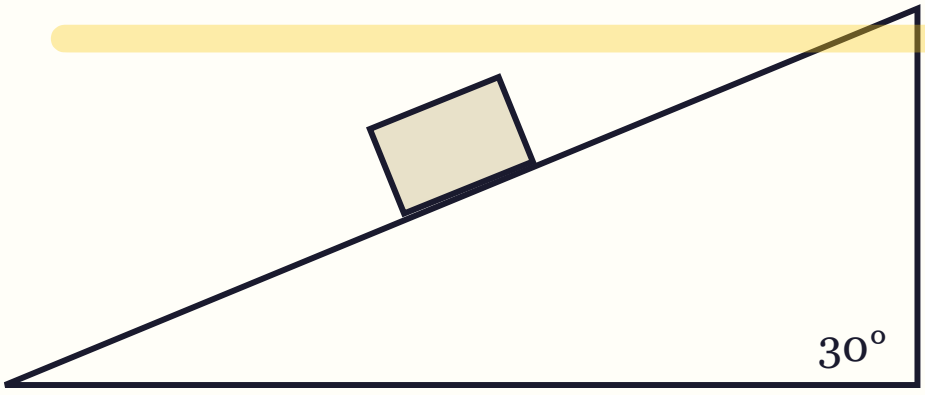
Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. (g = 10 m/s²)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

I took the weight as mg and forgot to resolve it along the incline.

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. (g = 10 m/s²)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

SOLUTION

Along the incline: a = g sin30 − μ g cos30 = 5 − 0.2 × 8.66 = 3.27 m/s², so the answer is (a) 3.3.

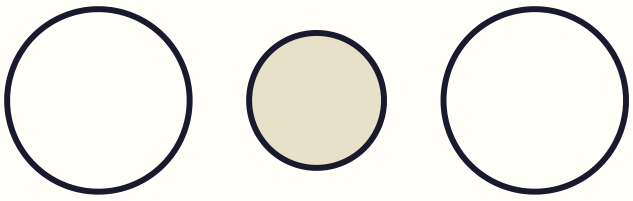
Fix: Resolve the weight along the incline before using friction.

Mistake 2

Moles of oxygen in CO_2

Mock · AITS 3 · Concept gap

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



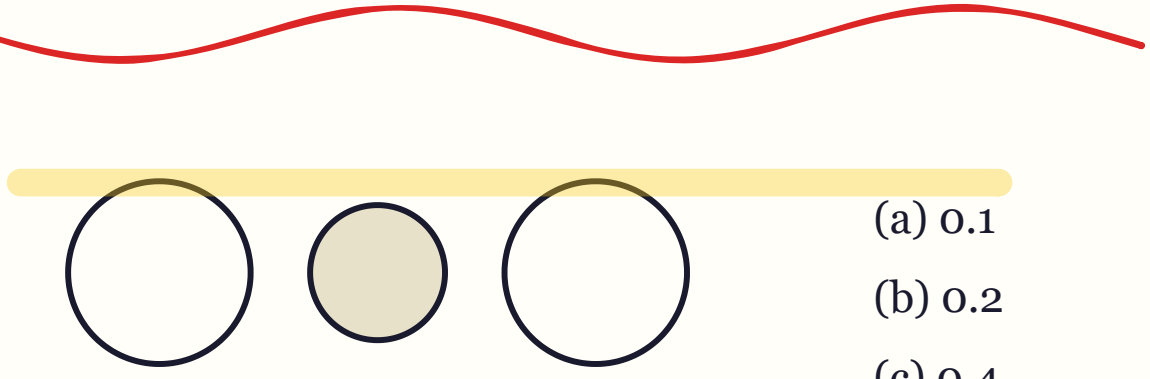
(a) 0.1

(b) 0.2

(c) 0.4

I took the weight as mg and forgot to resolve it along the incline.

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



(a) 0.1

(b) 0.2

(c) 0.4

SOLUTION

Along the incline: $a = g\sin30 - \mu g\cos30 = 5 - 0.2 \times 8.66 = 3.27$ m/s², so the answer is (a) 3.3.

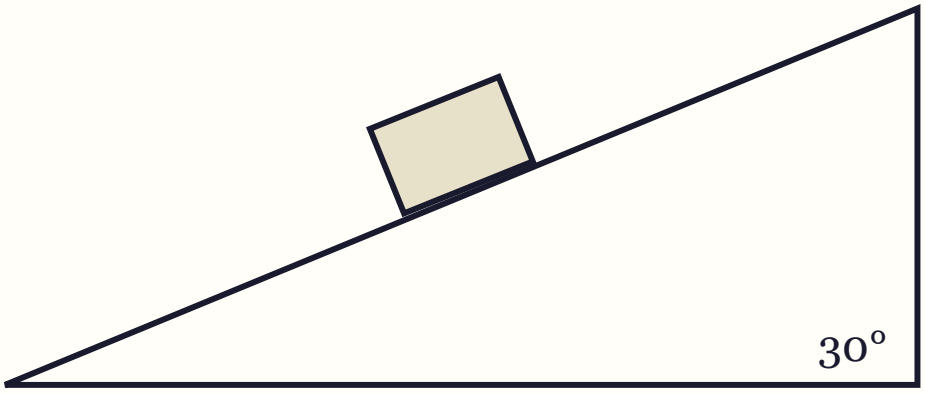
Fix: Resolve the weight along the incline before using friction.

Mistake 3

Projectile on an inclined plane

Coaching DPP · DPP 4 Q12 · Misread

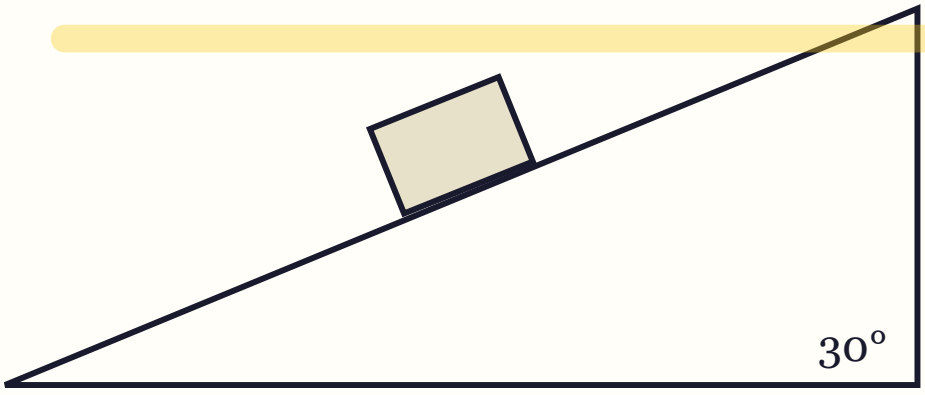
Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. (g = 10 m/s²)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

I took the weight as mg and forgot to resolve it along the incline.

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. (g = 10 m/s²)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

SOLUTION

Along the incline: a = g sin30 − μ g cos30 = 5 − 0.2 × 8.66 = 3.27 m/s², so the answer is (a) 3.3.

Fix: Resolve the weight along the incline before using friction.

Mistake 4

Relative velocity of two trains

PYQ · JEE Main 2023 · Time pressure

Ran out of time on the second part.

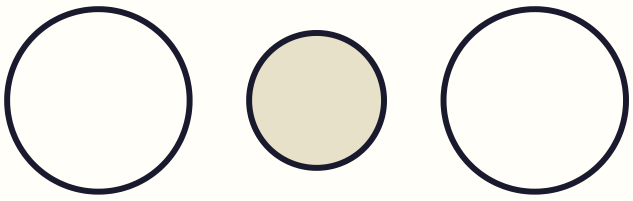
Fix: Resolve the weight along the incline before using friction.

Mistake 5

Stopping distance with friction

Coaching DPP · DPP 4 Q12

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



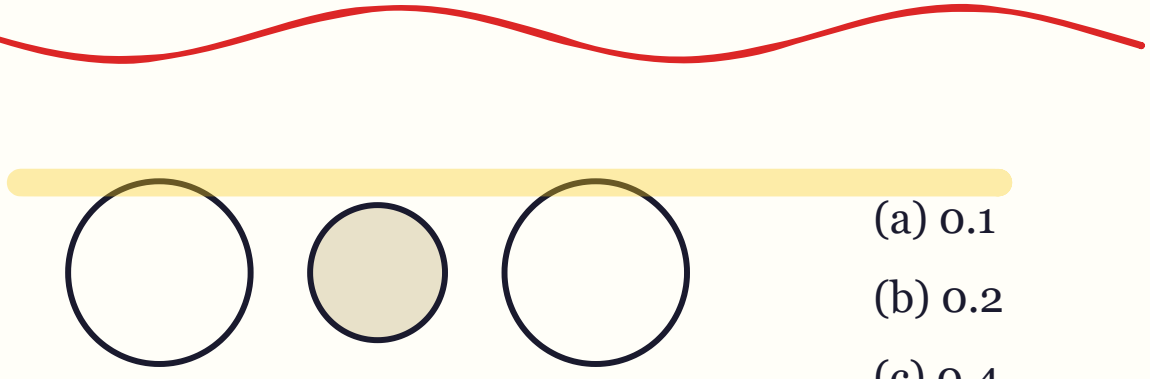
(a) 0.1

(b) 0.2

(c) 0.4

I took the weight as mg and forgot to resolve it along the incline.

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



(a) 0.1

(b) 0.2

(c) 0.4

SOLUTION

Along the incline: $a = g\sin30 - \mu g\cos30 = 5 - 0.2 \times 8.66 = 3.27\ m/s^2$, so the answer is (a) 3.3.

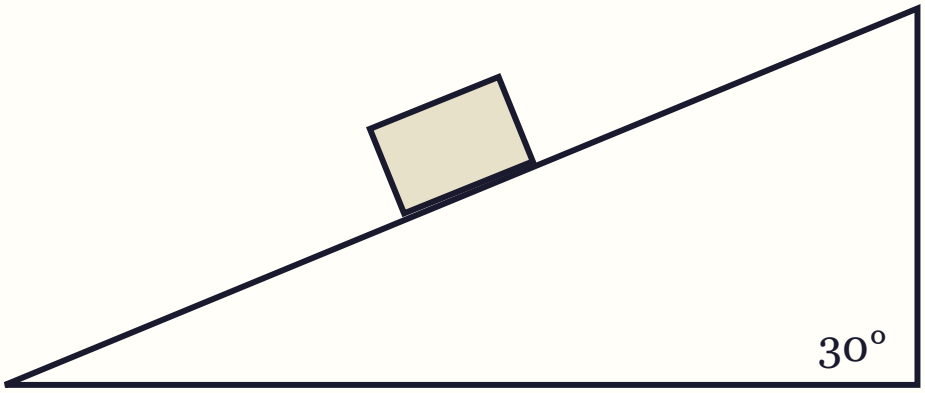
Fix: Resolve the weight along the incline before using friction.

Mistake 6

Free fall from a moving balloon

Coaching DPP · DPP 4 Q12 · Forgot formula

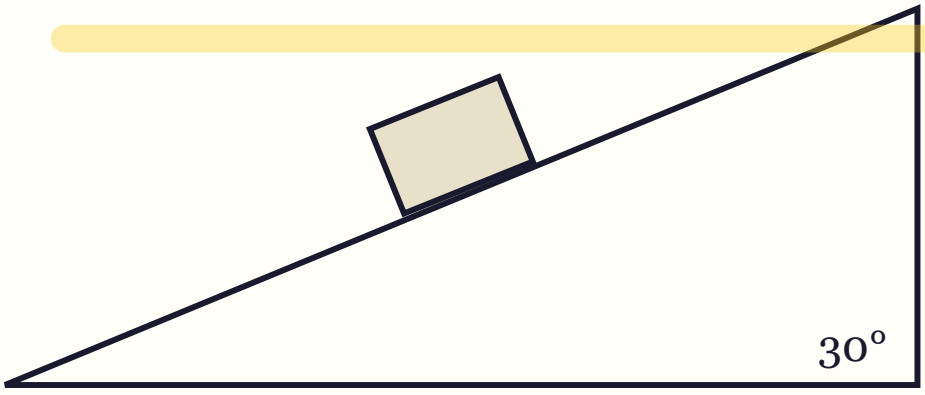
Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. (g = 10 m/s²)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

I took the weight as mg and forgot to resolve it along the incline.

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. (g = 10 m/s²)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

SOLUTION

Along the incline: a = g sin30 − μ g cos30 = 5 − 0.2 × 8.66 = 3.27 m/s², so the answer is (a) 3.3.

Fix: Resolve the weight along the incline before using friction.